CHE 205, Fall 2019, Exam 1

Instructor: Shafigh Mehraeen
Date: Fri, 09/27/2019

Note:

- Submit your spreadsheet (only one excel file) on blackboard before 10am.
- The exam is open-book and open-note. You are allowed to use internet as well; however, you are NOT allowed to talk to, text, or email anybody during the exam time.
- Make sure you save your file frequently as excel may crash every minute!
- Also keep a copy of your work just in case the submission fails to transmit the file.

	Ethylene	Ethane	Propane	n-Butane
A	3.867	3.933	3.977	3.844
B	584.146	659.739	819.296	909.650
C	-18.307	-16.719	-24.417	-36.146
z	0.1	0.3	0.4	0.2

Problem 1 (10 pts): Table above gives data for a mixture of four hydrocarbons. Included in the data are constants (A, B, and C) for the Antoine equation, a correlation that allows the calculation of the vapor pressure of pure components as follows:

$$log_{10}(P_i) = A_i - \frac{B_i}{C_i + T} \tag{1}$$

where P_i is the vapor pressure of component i (atm), T is the temperature (K), and A_i, B_i, C_i are Antoine coefficients. It is desired to find the fraction of the mixture in the vapor phase when the mixture is flashed at $T = 273$ (K), and total pressure of 18 (atm). Assume that z_i is the mole fraction of each component in the feed, $k_i = P_i/P$ is the equilibrium constant for component i, and P is the total pressure. Let α be the fraction of the feed flashed to the vapor phase. The governing equation for α is:

$$f(\alpha) = \sum_{i=1}^{4} \frac{z_i(1-k_i)}{1+\alpha(k_i-1)} = \frac{z_1(1-k_1)}{1+\alpha(k_1-1)} + \frac{z_2(1-k_2)}{1+\alpha(k_2-1)} + \frac{z_3(1-k_3)}{1+\alpha(k_3-1)} + \frac{z_4(1-k_4)}{1+\alpha(k_4-1)} = 0 \tag{2}$$

Set up a spreadsheet and solve for α using:

(a) Goal Seek (2 pts),

(b) Newton's method (8 pts).